Mini Projects using AI4Bharat & BharatGen

CS3235 | Undergraduate LLM / Multimodal AI Projects

Objective: Develop usable, end-to-end Indian-language AI applications using pretrained models, suitable frontends, and measurable evaluation. Suggested team size: 3–4 students.

1. Multilingual Voice-Based University Assistant

Application: Answer campus queries on fees, exams, academic policies and admissions in Hindi, Kannada or Tamil, with voice input and output.

Models / data: AI4Bharat IndicConformer, IndicTrans2, Indic Parler-TTS; BharatGen Param1

Concepts: ASR, translation, RAG, text-to-speech, agents | **Evaluation:** Word Error Rate (WER), grounded answer accuracy

2. AI Farmer Advisory Assistant

Application: Provide Indian-language guidance for crop, irrigation and fertilizer queries, backed by curated agricultural sources.

Models / data: BharatGen AgriParam; AI4Bharat IndicTrans2; agricultural advisory documents

Concepts: Domain LLM, document retrieval, RAG | **Evaluation:** Groundedness, factual correctness, domain QA

3. Indian Document Understanding Tool

Application: Upload an Indian-language notice, form or government document; extract fields, summarize and answer document questions.

Models / data: BharatGen Patram vision-language model; AI4Bharat IndicTrans2

Concepts: Computer vision, OCR/document VQA, multimodal LLM | **Evaluation:** Field-extraction accuracy, QA accuracy

4. Hinglish Smart Writing Assistant

Application: Convert Roman-script Hinglish into native script, improve phrasing and optionally translate into English.

Models / data: AI4Bharat IndicXlit, IndicTrans2; BharatGen Param1; Aksharantar dataset

Concepts: Transliteration, translation, prompting | **Evaluation:** Character Error Rate, word accuracy, translation quality

5. Multilingual Lecture-to-Notes & Quiz Generator

Application: Transcribe a recorded lecture, create clear notes and five MCQs, and optionally translate the notes.

Models / data: AI4Bharat IndicConformer or BharatGen speech model; Param1; IndicTrans2

Concepts: ASR, summarization, quiz generation | **Evaluation:** WER, factual consistency, MCQ quality

Assessment Rubric

Criterion
Problem definition & relevance
Model / dataset selection
Working end-to-end application
GenAI / RAG / agentic integration
Quantitative testing & error analysis
Demo, documentation & repository

Require 30–50 representative test cases, a failure analysis, a usable frontend (e.g., Gradio/Streamlit/React), and a GitHub repository.

Official Starting Points

AI4Bharat IndicTrans2: https://github.com/AI4Bharat/IndicTrans2

AI4Bharat IndicConformer: https://github.com/AI4Bharat/IndicConformerASR

AI4Bharat Indic Parler-TTS: https://huggingface.co/ai4bharat/indic-parler-tts

BharatGen organization: https://huggingface.co/bharatgenai

AI4Bharat Aksharantar: https://github.com/AI4Bharat/IndianNLP-Transliteration

Implementation note: Confirm specific checkpoint availability, licensing, language coverage, GPU requirements, and suitability before assigning a model. Dataset benchmarks are not substitutes for task-specific evaluation.